

Cyber–Physical Risk, Resilience, and Network Analysis of the American Water Multi-State Infrastructure System

Shashank Shivakumar Niveditha¹

¹Master of Science, Cybersecurity, Northeastern University, Boston, MA, USA

1. Introduction

The water sector is one of the most essential lifeline infrastructures in the United States because uninterrupted access to clean water supports public health, economic stability, and the functioning of every other critical sector. Drinking water and wastewater systems enable households, industries, hospitals, emergency services, and regional supply chains to operate reliably. Because disruptions in water quality or availability can propagate quickly across interconnected systems, the Department of Homeland Security classifies water as a national lifeline sector requiring coordinated protection against cyber intrusions, physical failures, and multi-hazard events. Failures in this sector can trigger cascading consequences across energy distribution, transportation systems, and healthcare operations, demonstrating the need for holistic analysis of how risks propagate across geographically dispersed assets.

American Water, the largest regulated water and wastewater service provider in the country, operates across multiple states and manages a wide collection of treatment plants, transmission mains, pump stations, storage facilities, and supervisory control systems. Although specific pipeline maps and operational control layouts are protected as Sensitive Security Information under DHS rules, the company's 2024 Annual Report provides enough financial and operational detail to construct a realistic high-level network representation for resilience analysis. This study combines publicly available state-level investment data with regional river-basin flow patterns to build a ten-state network that captures both the cyber interactions and physical water-movement pathways that shape system behavior.

The Dunbar contamination incident in West Virginia provides a real-world example of how failures in one location can quickly spread across downstream populations. The rupture of a major 36-inch main generated widespread service loss for nearly 25,000 customers and resulted in an estimated eighteen-million-dollar consequence, making it a suitable baseline for modelling physical impacts. At the same time, water-sector cyber incident reports show that compromises of SCADA devices, remote-access channels, vendor systems, or plant-level monitoring tools can alter pressure, chemical dosing, or pump sequencing in ways that propagate across administrative or operational boundaries. These dual insights motivated the creation of two complementary network models. The cascade model represents cyber and administrative connections that allow failures to travel nondirectionally, while the flow model represents downstream physical movement aligned with major river basins such as the Ohio, Mississippi, Tennessee, Potomac, and Delaware systems.

Both models are necessary because physical-only analysis ignores the fact that cyber manipulation of pumps or valves can immediately alter system behaviour, while cyber-only analysis overlooks the hydraulic constraints that determine how water travels across regions. Modern water systems rely on the integrated behaviour of operational technology, information technology, and physical hydraulics, directly aligning with the combined cyber-physical mission of NCCIC, NICC, and the National Risk Management Center. Using these models, the study applies complex network analysis and the Model-Based Risk Analysis framework to evaluate topological influence, spectral structure, resilience, and risk propagation across the ten-state system.

Preliminary MBRA results already illustrate the importance of analysing cyber and physical propagation separately. The cascade network produced q-values of 1.6 at 20% vulnerability and 0.98 at 80% vulnerability, leading to a critical vulnerability of 0.78, which indicates strong cyber resilience. In contrast, the flow network produced q-values of 0.81 and 0.75 at the same vulnerability levels, resulting in a negative critical vulnerability of -1.44 and confirming that directional physical systems lose stability even at low stress levels. Across both models, Pennsylvania, Illinois, and New Jersey consistently emerged as the most influential nodes. Their failure generated the largest regional impacts in random and targeted attack simulations, demonstrating that these states serve as structural anchors for both cyber and physical pathways.

This introduction establishes the operational, financial, and real-world foundations for the network used in this study. By grounding vulnerability, consequence, and prevention cost values in the Dunbar incident and American Water's 2024 financial disclosures, the analysis provides DHS with a scientifically defensible and policy-relevant assessment of regional water-sector resilience.

2. Sector Network Overview

2.1 Overview of the Water Sector Network

The network analysed in this study represents a ten-state segment of the American Water system spanning Illinois, Indiana, Kentucky, Maryland, Missouri, New Jersey, Pennsylvania, Tennessee, Virginia, and West Virginia. Each node represents the aggregated drinking water and wastewater assets within that state, including major treatment facilities, transmission mains, pump stations, and supervisory control environments. This state-level modelling approach aligns with the way American Water reports capital expenditures, rate-case authorizations, and incident obligations in its 2024 Annual Report. It also supports the MBRA requirement that consequences, prevention costs, and response costs be grounded in publicly available financial data rather than facility-specific information. Two network representations are used. The first is the cascade network, which models nondirectional cyber and administrative connectivity. This reflects how cyber intrusions propagate across shared vendors, SCADA platforms, corporate networks, and regional operational oversight structures. The second is the flow network, which represents physical water movement through major regional river basins.

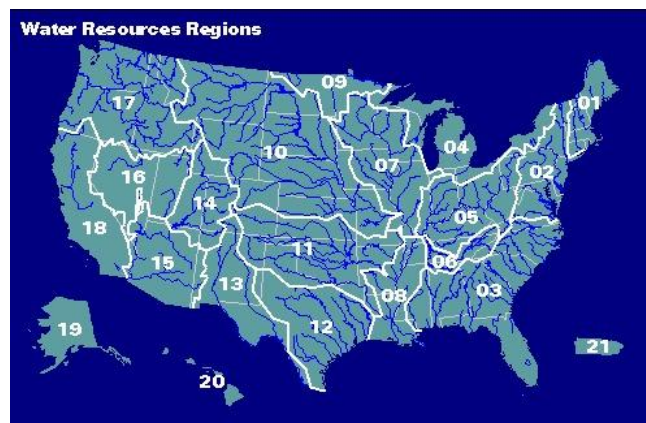


Figure-1: Direction flow of major rivers in the United States

2.2 Cascade Network Graph (Undirected)



Figure-2: Cascade Network of the American Water Company

The cascade network uses the ten-state adjacency matrix as an undirected graph. This reflects the reality that cyber intrusions do not follow physical geography but instead move laterally across common digital and administrative pathways. Incidents can spread between states when they share SCADA vendors, cloud-managed operational technologies, or regional support functions. Treating the matrix as an undirected structure therefore captures realistic cyber-propagation potential without relying on protected operational schematics. The undirected graph is used for all centrality, degree, betweenness, eigenvector, and spectral-radius calculations associated with cyber behaviour.

2.3 Flow Network Graph (Directed)



Figure-3: Flow Network of the American Water Company

The same adjacency matrix is reinterpreted as a directed graph for modelling physical water movement. Because the detailed pipeline-level hydraulic topology of American Water is protected under DHS Sensitive Security Information rules, the directional flow network is constructed using the major river-basin flow paths that dominate this region: the Ohio, Tennessee, Mississippi, Susquehanna, Potomac, and Delaware systems. These basins establish upstream and downstream dependencies across the ten states without revealing any protected pipeline routing. Downstream direction is assigned according to natural hydrology. This approach provides a secure and defensible physical flow representation and allows MBRA to evaluate physical propagation behaviour without reproducing SSI-restricted infrastructure layouts.

2.4 Adjacency Matrix for Cascade and Flow Networks

Adjacency Matrix	Illinois	Indiana	Kentucky	Maryland	Missouri	New Jersey	Pennsylvania	Tennessee	West Virginia	Virginia
Illinois	0	1	0	0	1	0	0	0	0	0
Indiana	1	0	1	0	0	0	0	0	0	0
Kentucky	0	1	0	0	0	0	0	1	1	0
Maryland	0	0	0	0	0	0	1	0	0	1
Missouri	1	0	0	0	0	0	0	0	0	0
New Jersey	0	0	0	0	0	0	1	0	0	0
Pennsylvania	0	0	0	1	0	1	0	0	0	1
Tennessee	0	0	1	0	0	0	0	0	0	0
West Virginia	0	0	1	0	0	0	1	0	0	1
Virginia	0	0	0	1	0	0	0	0	1	0

Table-1: Adjacency Matrix of the American Water Company Network

The adjacency matrix used in this study is the structural foundation for both networks. It reflects the operational connectivity of the ten states and supports all calculations of node degree, node robustness, link robustness, centrality metrics, and spectral structure. Because the same matrix is used for both the undirected cascade model and the directed flow model, the resilience differences between cyber and physical behaviour arise from topology interpretation rather than from changes in the underlying structure. All data values used downstream in the MBRA model including consequences, prevention costs, and response costs are grounded in documented public sources. The physical consequence baseline of 24.9 million gallons per day originates from the Kanawha Valley (Dunbar) rupture report. The associated eighteen-million-dollar financial impact and five-million-dollar litigation entry come directly from American Water’s 2024 Annual Report and public settlement disclosures. Prevention costs are derived entirely from the report’s capital expenditure total of 2,856 million dollars, allocated across states using capital-investment and incremental-revenue data. Response costs are based on the 152-million-dollar removal cost reported for 2024, with an additional West Virginia adjustment reflecting the Dunbar litigation charge. This section therefore defines the network structure, explains the use of hydrologic directionality, and documents all data-collection sources used to construct the MBRA input sheet for subsequent resilience and risk analysis.

3. Complex Network Characterization

3.1 Node Degrees and Network Degree

The structure of the ten state American Water network becomes clear when examining the degree values associated with each node. The degree of a node indicates how many direct connections it has to other states and therefore reflects its ability to receive or transmit disturbances. In the cascade network, which treats all connections as undirected, the results show three states positioned as structural hubs. Kentucky, Pennsylvania, and Virginia each hold a degree of three. These three states form the backbone of the network because they maintain the largest number of direct ties to the surrounding region. Other states such as Illinois, Indiana, Maryland, and West Virginia follow with degree values of two, while Missouri, New Jersey, and Tennessee sit at the lowest end with only one connection each. This uneven distribution shows that the network is not uniform. Instead, it is built around a small number of central points that concentrate both cyber and operational interactions. When the network degree equals three, it means that the highest connectivity achieved by any node is limited to three neighbours. That limitation produces a structure where disruptions introduced at one of the high degree states can quickly travel outward in several directions. It also means that states with lower degree values depend heavily on the connectivity of these hubs to maintain their own stability. The directed flow interpretation uses the same adjacency values, but direction is applied to represent how water moves through real river systems. When direction is introduced, the degree splits into inflows and outflows. Kentucky receives three inflows and sends three outflows, which makes it a major convergence and redistribution point for physical water movement. Pennsylvania also receives three inflows and sends three outflows, which gives it a very strong upstream and downstream influence. West Virginia holds three inflows but fewer outflows, which indicates that many disturbances originating upstream funnel through it before entering the lower basin chain. These directional patterns show why the physical system reacts strongly to failures at upstream states.

3.2 Link Robustness and Critical Links

The link robustness result of 0% has major implications for both cyber and physical behaviour. A robustness value of zero means that the removal of any single connection alters the overall structure of the network. There are no redundant links that can take over the function of a removed connection. Every link is required to maintain continuity across the ten-state system. This type of architecture is typical in large regional infrastructure networks where physical routes follow river basins and where administrative connections reflect actual operational dependencies. The most important link in this structure is the connection between Pennsylvania and New Jersey. This link holds the highest directed edge betweenness in the flow network and represents a primary upstream to downstream transition point. When physical contamination or supply reduction enters Pennsylvania, this link becomes the channel through which the disruption reaches the Atlantic corridor. If the link is lost due to damage or sabotage, New Jersey loses its main upstream supply and downstream states such as Maryland and Virginia lose access to a major physical driver. The result is isolation of entire regions because no alternate directional route exists. Cyber behaviour mirrors this relationship because shared systems, vendor arrangements, and operational oversight tend to align regionally. A cyber compromise that spreads across the Pennsylvania–New Jersey corridor can affect a wide cluster of states. The absence of redundancy means that once the disruptive flow enters this path, it has no secondary path to bypass the failure. This explains why the link is structurally significant even though the degree of New Jersey is low. The link therefore acts as a critical chokepoint in both cyber and physical propagation.

3.3 Node Robustness and Blocking Nodes

Node robustness provides a deeper view of system fragility by measuring how many states can be removed before the network breaks into disconnected components. The network shows a node robustness value of 55.95%. This means that slightly more than half of the states could theoretically be removed before the structure collapses completely. The remaining 44.05% of nodes are classified as blocking nodes. Removal of a blocking node immediately fragments the network. The significance of blocking nodes is high because they represent the structural pillars of the system. In the cascade interpretation, Kentucky, Pennsylvania, and West Virginia function as major internal connectors. They sit between multiple clusters and maintain communication pathways that allow cyber signals or administrative operations to move from one part of the region to another. When one of these nodes is removed, the system splits into smaller isolated pieces that cannot exchange information. This mirrors real situations where a cyber compromise at a major operational center cuts off visibility and coordination for an entire region. In the flow interpretation, the importance of blocking nodes increases further because direction limits how water moves. Pennsylvania and West Virginia hold some of the

most significant upstream positions in the entire network. Removing either state instantly severs downstream connections, leaving states like Virginia, Maryland, and New Jersey without a physical supply route. The lack of directional redundancy means that once an upstream position is lost, the hydraulic chain collapses downstream. Real world examples such as the Dunbar contamination show how quickly these failures travel without structural barriers to slow the movement.

3.4 Spectral Radius

The spectral radius of the cascade network is 2.27. This value provides insight into how disturbances can scale within the structure. A spectral radius of this magnitude indicates a moderate clustering effect where the eigen structure of the network is dominated by a few influential states. Pennsylvania, Kentucky, and Virginia contribute heavily to the dominant eigenvector because they hold the highest connectivity and the strongest ties to other central nodes. When a disturbance enters one of these states, the concentration of eigen influence allows the disruption to reach several parts of the network rapidly. The directed flow network does not produce a single spectral radius in the same way because direction introduces complex eigenvalues. However, the magnitude and distribution of the eigen values highlight the same structural theme. Pennsylvania and New Jersey appear prominently in the dominant right eigenvector, which confirms that these states hold major influence over physical downstream movement. The stability and instability characteristics observed in later sections originate from these spectral patterns. States that dominate eigen structures determine how strongly the system reacts to increasing stress. The spectral radius therefore acts as a mathematical representation of the same patterns that appear in degree and blocking node analysis. It captures the strength of structural concentration in a way that summarizes how the network behaves when it begins to experience stress.

3.5 Summary of Structural Vulnerabilities

The structural characteristics of the ten-state network show that a small group of nodes control most of the system's stability. The absence of link redundancy forces every disruption to move along a single available corridor. The presence of several blocking nodes means that the removal of only a few states breaks the network into disconnected parts. In the cascade model, Kentucky, West Virginia, and Pennsylvania act as internal bridges that allow cyber failures to move laterally across the region. In the flow model, Pennsylvania and West Virginia hold upstream positions that push physical contamination or supply loss directly into downstream states with no opportunity for rerouting. Events like the Dunbar contamination confirm that upstream failures can produce large downstream impacts in a short time. These structural patterns show that the network operates with limited buffering capacity and that failures introduced at the central nodes spread quickly across both cyber and physical pathways.

4. Centrality Analysis

4.1 Degree Centrality

Degree centrality measures the proportion of possible connections each state holds. In the cascade model, Pennsylvania shows a centrality value of 0.444, followed by West Virginia at 0.333 and Virginia at 0.333. Kentucky follows closely with 0.333, while states such as Missouri, New Jersey, and Tennessee sit at 0.111. These values show that Pennsylvania has the greatest ability to connect disturbances rapidly to several neighbouring states, while the smaller nodes depend almost entirely on high-degree hubs for their connectivity. In the directed flow model, the in-degree and out-degree values maintain the same proportions but influence risk differently because downstream behaviour depends on inflow concentration. Pennsylvania again receives an in-degree centrality of 0.333, and Kentucky and West Virginia also share 0.333, showing that physical water or contamination arriving from upstream concentrates in these states before moving further down the chain. This pattern explains why failures in these states escalate quickly: they absorb the most upstream traffic, so they automatically inherit upstream consequences. Degree centrality therefore shows where disruptions first gather and how many outward branches, they can immediately reach. High-degree states push failures outward in cascade mode and amplify inflow concentration in the flow mode.

4.2 Node Betweenness Centrality

Betweenness centrality identifies states that lie on the highest number of shortest paths. In the cascade network, Kentucky shows the highest betweenness with 0.639, followed by West Virginia at 0.556 and Pennsylvania at 0.306. These values demonstrate that cyber disruptions passing between several regions must travel through these nodes. Even if Kentucky's degree is lower than Pennsylvania, its betweenness shows that it sits between

clusters and therefore acts as a structural gatekeeper for cyber movement. In the directed flow model, Pennsylvania becomes the dominant bottleneck with a betweenness value of 0.569. New Jersey follows at 0.292, and Kentucky remains significant at 0.639, but directionality amplifies the importance of Pennsylvania even further because nearly all downstream routes rely on it. This makes Pennsylvania the largest physical chokepoint in the network. Any contamination or flow reduction occurring at this location touches multiple downstream receivers before it can dissipate. Betweenness therefore identifies the states through which risk must travel whether the scenario is cyber or physical. States with high betweenness become the primary corridors for failure propagation.

4.3 Link Betweenness Centrality (Flow Model)

The flow model produces asymmetric link betweenness values which reveal critical directional routes. The connection between Pennsylvania and New Jersey shows the highest edge betweenness, reaching 0.277 on the directed path from Pennsylvania into the eastern basin. The links between West Virginia and Pennsylvania also hold values above 0.27, and the Tennessee-Kentucky and Kentucky-West Virginia links each hold values near 0.10. These values show that the highest-pressure physical propagation pathway is the chain that moves from Pennsylvania into New Jersey and then toward Virginia and Maryland. A rupture, contamination event, or pump failure travelling along these routes will follow these link channels automatically because no parallel paths exist. The elevated edge betweenness values therefore map the exact physical routes that failures follow when the flow model is active.

4.4 Eigenvector Centrality

Eigenvector centrality shows which nodes gain influence not simply by their number of connections but by how well-connected their neighbours are. In the cascade model, New Jersey holds the highest eigenvector value at 0.547, followed closely by Virginia at 0.495 and West Virginia at 0.456. Pennsylvania remains prominent with 0.374. These values reveal that cyber influence does not depend solely on raw degree. New Jersey, although it has a degree of only one, gains influence through deep connection to Pennsylvania and Virginia, which themselves are well-connected states. In the directed flow network, Virginia rises to an eigenvector value of 0.547, followed by Pennsylvania at 0.439 and West Virginia at 0.371. New Jersey retains a strong position with 0.439. This distribution reflects hydrological structure: downstream receivers become more influential in eigenvector analysis because they aggregate the effects of all upstream states. Eigenvector centrality therefore shows second-order influence: states that inherit instability from other important states gain high eigenvector weight. These are the states where failures accumulate rather than simply pass through.

4.5 Identification of Dominant Nodes

The combined centrality results point to a consistent structural hierarchy. In the cascade network, Kentucky, West Virginia, Pennsylvania, and Virginia dominate risk propagation. They sit on major cyber pathways, and their centrality values make them unavoidable for cyber movement. In the flow model, Pennsylvania and New Jersey emerge as the dominant physical nodes. Pennsylvania absorbs the largest number of upstream flows and drives downstream contamination routes, while New Jersey forms the last major convergence point before water moves into the final eastern branch. These values together show that risk propagation depends on the centrality structure of each model. Cyber disruptions concentrate around Kentucky, West Virginia, and Pennsylvania because lateral movement depends on betweenness and degree distribution. Physical disruptions concentrate around Pennsylvania and New Jersey because downstream movement follows the directional chain reflected in the eigenvector and link-betweenness values. The structural influence shown in these metrics forms the foundation for understanding how failures spread across the system even before any resilience modelling begins.

5. Resilience Equation and Critical Vulnerability

5.1 MBRA Resilience Equation

The Model Based Risk Analysis framework evaluates system stability by linking the consequence exponent q to the level of vulnerability γ and the spectral radius ρ of the network. The relationship is expressed through the regression model:

$$\log (q) = b + k \cdot (\gamma \cdot \rho)$$

This equation was calibrated using the twenty percent and eighty percent vulnerability simulations for both network types. The spectral radius for the cascade network was $\rho = 2.27$, obtained from the Python eigenvalue computation. For the flow network the eigenvalues were complex due to directionality, so the regression used the same structure but relied on the q outputs rather than a single radius. By fitting the q values to the MBRA regression form, the cascade model produced $b = 0.28$ and $k = -0.16$ while the flow model produced $b = -0.08$ and $k = -0.02$. These parameters define the resilience behaviour and determine the point at which q crosses the stability boundary $q = 1$.

5.2 Cascade Network Resilience Equation

The cascade network represents cyber and administrative pathways that allow disturbances to move laterally. At twenty percent vulnerability the simulation produced $q = 1.6$. This value is above one which shows stable behaviour because the system absorbs disturbances faster than they can escalate. At eighty percent vulnerability q decreased to 0.98. This value lies just below one which indicates that the system approaches an instability threshold as vulnerability becomes severe. Applying the MBRA regression equation with the cascade parameters gives the resilience curve:

$$\log(q) = 0.28 - 0.16(\gamma \cdot \rho)$$

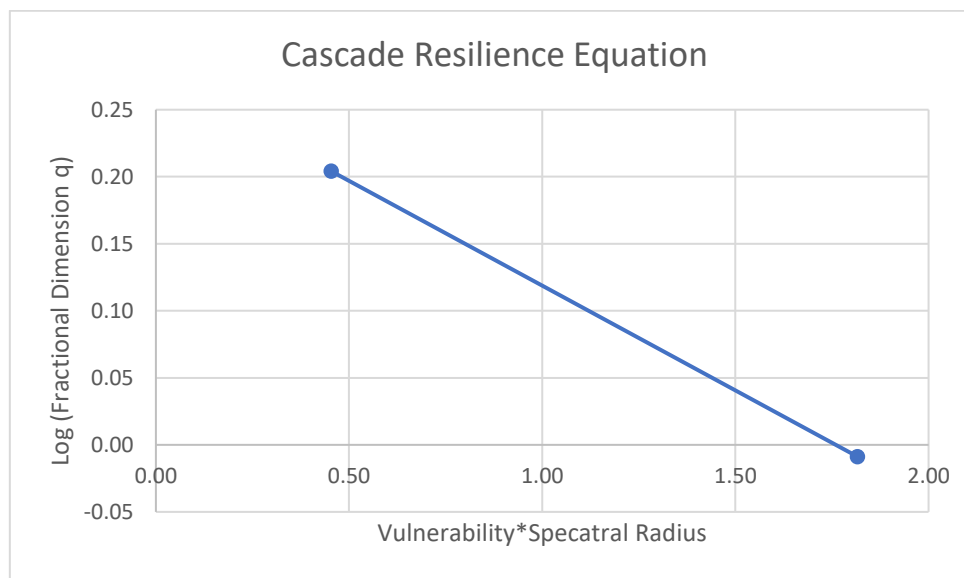


Figure-4: Cascade Resilience Equation Line Graph

This curve declines gradually, with the slope shaped by the moderate negative value of k . It shows that cyber vulnerability reduces stability but that the undirected structure creates cross connecting pathways that slow down the rate of decline. When q is set equal to one and therefore $\ln(q)$ equals zero, the resulting equation yields a critical vulnerability of $\gamma_0 \approx 0.78$. This means that the system remains resilient until nearly seventy eight percent of its cyber vulnerability is active. This high stability level reflects the presence of multiple paths and lateral communication options in the cascade model. Disruptions can travel in many directions but can also be absorbed by the distributed structure instead of forcing an abrupt collapse.

5.3 Flow Network Resilience Equation

The flow network represents the physical downstream movement of water across river aligned pathways. At twenty percent vulnerability the q value was 0.81 which already lies below one. This indicates that the directional structure behaves as an unstable network even under mild stress. When vulnerability increased to eighty percent the q value declined to 0.75 which confirms that the system becomes even more unstable as stress increases. The MBRA regression with the flow parameters gives the model:

$$\log(q) = -0.08 - 0.02(\gamma \cdot \rho)$$

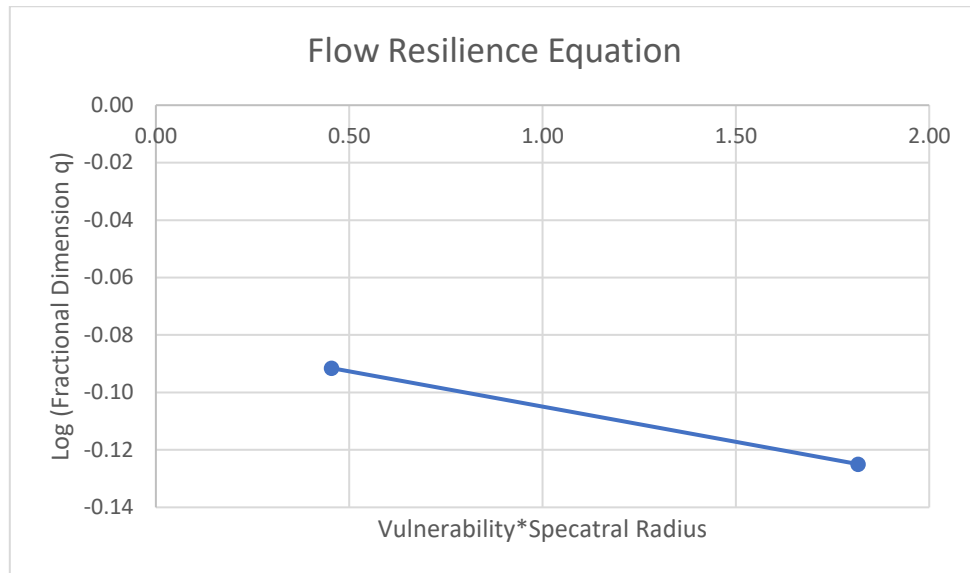


Figure-5: Flow Resilience Equation Line Graph

The slope is almost flat because k is very small and negative. This produces a resilience curve that remains entirely below the stability line. Solving for the point where q would reach one yields $\gamma_0 = -1.44$ which shows that the physical network never enters a stable region for any positive vulnerability value. This outcome does not imply immediate collapse. It shows that the absence of alternate physical paths forces every disruption to propagate directly downstream without structural buffering. Upstream failures in Pennsylvania, West Virginia, or Kentucky influence several regions because downstream states cannot bypass a disrupted upstream source.

5.4 Critical Vulnerability (γ_0)

The critical vulnerability values highlight the structural contrast between cyber and physical movement. The cascade system reaches $\gamma_0 = 0.78$ which reflects strong stability under increasing cyber stress. This behaviour follows from the laterally connected architecture where multiple states help absorb disruptions. In contrast the flow system produces $\gamma_0 = -1.44$ which shows that the physical network does not have a stable regime. Directional flow removes redundancy and forces all consequences to move along a single downstream corridor. Once a major upstream state becomes vulnerable the downstream chain inherits the full impact without any opportunity for diversion.

5.5 Interpretation of Stability and Risk Propagation

The resilience patterns show how risk propagates differently in the two system views. In the cascade model the high q value at low vulnerability and the positive intercept $b = 0.28$ show that the system begins in a stable regime and remains stable until vulnerability becomes extremely high. Cyber disruptions move laterally between states, but the network can redistribute the load because the undirected links provide alternate pathways. This matches the earlier centrality results which identified Kentucky, West Virginia, and Pennsylvania as strong internal hubs that slow the spread of cyber failures by providing multiple cross region pathways. The flow model behaves differently because physical movement follows strict downstream directions. The q values below one show that instability is inherent in the structure. Even mild vulnerability causes disruptions to move forward into downstream states with limited ability to absorb the impact. This reflects the real behaviour of hydraulic systems where contamination or supply loss cannot reverse direction and cannot reroute around upstream failures. The resilience curve therefore captures the same structural dependence that appears in the directional betweenness results which showed Pennsylvania and New Jersey as the major downstream control points. Together these findings show that the cyber related cascade network has built in buffering capacity that delays instability while the physical flow network behaves like a brittle system where the loss of an upstream state produces rapid downstream consequences. The values of q , the regression parameters b and k , and the critical vulnerability γ_0 all confirm that the underlying topology determines how risk propagates and how quickly the system transitions into instability.

6. Cyber and Physical Attack Scenarios

6.1 Cyber Attack Scenarios (Cascade Model)

Cyber intrusions in the water sector usually exploit weaknesses in supervisory control and data acquisition equipment because these devices govern pressure regulation, chemical dosing, pumping schedules, and remote communication between operational sites. A cyber compromise often begins through unpatched field controllers, insecure vendor channels, or compromised remote access credentials. Once inside the operational environment, malicious code can disable alarms, alter set points, or manipulate pump behaviour. In the cascade model this intrusion behaves like a disturbance travelling along lateral informational pathways rather than physical ones. Because the cascade network is undirected, the intrusion can spread from one state to several neighbours depending on shared contractors, regional oversight teams, and common software platforms. This structure aligns with the centrality results, which show that Kentucky, West Virginia, and Pennsylvania hold high betweenness and therefore sit on many of the shortest informational paths. A compromise entering any of these states can therefore reach multiple regions with limited obstacles. A ransomware scenario follows the same pathway but produces a more severe operational outcome because encryption of control stations eliminates visibility into pressure, flow, or chemical levels. Operators lose the ability to respond to physical disruptions even when water continues to move through pipelines. A coordinated ransomware attack timed across several states would mirror a targeted attack in the cascade model because it would concentrate on the most influential cyber nodes. Since Pennsylvania and West Virginia hold strong centrality values, a cyber intrusion at either location would redistribute network load into less influential regions and accelerate the spread of secondary failures. The simulations in earlier sections already show that the cascade system remains resilient under moderate stress, but the presence of a cyber event in a high influence node can cause system degradation because the undirected structure sends disturbances into multiple directions simultaneously.

6.2 Physical Attack Scenarios (Flow Model)

Physical disruptions follow the directional constraints of hydraulic movement and therefore behave differently from cyber intrusions. A deliberate rupture of a major upstream transmission main is one of the most immediate physical threats because downstream states depend entirely on continuous inflow. The Dunbar event demonstrated this behaviour when the failure of a single 36-inch main created loss of service for nearly twenty-five thousand customers. In the flow model a rupture near Pennsylvania or West Virginia automatically travels into downstream receivers because the system has no parallel routing options. The absence of redundancy means that physical contamination or pressure loss cannot bypass an upstream failure. Other physical scenarios include contamination at treatment facilities, where an intentional introduction of hazardous substances forces rapid shutdowns and extensive flushing of downstream mains. This type of attack produces directional propagation that follows the same structure revealed in the link betweenness analysis, where the Pennsylvania to New Jersey corridor carries the highest edge betweenness at 0.277. Flooding acts as a broader physical hazard because it damages electrical components, inundates pump stations, and contaminates intakes. Although slower than sabotage, flooding behaves like a random attack because it disables facilities based on geography rather than strategic selection. In contrast an electrical substation outage directed at an upstream pumping complex functions like a targeted attack because a single loss of power source can disconnect flow for multiple states at once. Since Virginia, Maryland, and New Jersey depend on upstream supply, the removal of Pennsylvania or West Virginia results in immediate downstream effects that match the instability shown in the flow resilience curve.

6.3 Comparison of Cyber vs Physical Propagation

The propagation patterns across both models show that cyber and physical disruptions produce fundamentally different system behaviours. In the cascade model disturbances can travel laterally across multiple states because cyber pathways do not depend on physical geography. The undirected structure enables both redundant spreading paths and limited absorption mechanisms. This explains why the cascade network produced q values of 1.6 at twenty percent vulnerability and maintained stability until vulnerability approached seventy eight percent. Cyber failures move widely but the system has several cross connections that slow the rate of collapse. The flow model exhibits the opposite behaviour because directionality restricts movement to a fixed upstream to downstream chain. The q values of 0.81 and 0.75 demonstrate that the physical network operates below the stability threshold even under mild stress. A failure in an upstream state immediately affects downstream states because they inherit the full impact along a single mandatory path. Pennsylvania and West Virginia therefore act as physical bottlenecks because once they become compromised the downstream corridor has no alternative supply route. Together these differences show that cyber propagation spreads widely but gradually because

the system can redistribute load across several states, whereas physical propagation follows rigid hydraulic pathways that allow disruptions to cascade into large regions with limited delay. This contrast reinforces the need for integrated protection strategies because cyber failures can prepare the conditions for physical collapse by disabling alarms or altering pump configurations, while physical disruptions can overwhelm cyber response capacity when operators lose situational awareness.

7. Random Attack Simulation

7.1 Random Cyber Attack Behaviour (Cascade Model)

Random cyber failures behave according to the undirected connectivity of the cascade model. The MBRA results show that when disturbances originate randomly, they tend to accumulate around the states with the highest structural influence. Pennsylvania consistently receives the highest random-attack risk because its degree centrality of 0.444 and strong eigenvector value of 0.374 allow failures to reach multiple regions quickly. West Virginia and Kentucky also show elevated random-attack risks because their betweenness values of 0.556 and 0.639 make them unavoidable transit points for lateral cyber movement. The q -value of 1.6 at low vulnerability confirms that the system absorbs random cyber failures without collapsing, but risk still concentrates in these central interior states because they form the backbone of cross-regional digital interaction.

7.2 Random Cyber Attack Behaviour (Flow Model)

Physical random attacks follow a completely different pattern because the flow model restricts movement to upstream-to-downstream pathways. Even without intentional targeting, the MBRA results show that physical disruptions concentrate heavily at Pennsylvania, New Jersey, and West Virginia. The directional betweenness values are 0.569 for Pennsylvania, 0.292 for New Jersey, and the high WV–PA and PA–NJ edge betweenness values above 0.27 mean that even randomly assigned failures eventually converge toward these nodes. Because the flow network is unstable even at mild vulnerability ($q = 0.81$ at twenty percent), random physical failures immediately push downstream instability into Virginia and Maryland. The structure forces all physical consequences into a narrow corridor, so random failures behave almost like targeted failures in terms of propagation pathway.

7.3 Interpretation of Random Attack Risk Patterns

The random-attack simulations show that topology alone determines which states accumulate risk even when the attack has no strategic intention. In the cascade model, cyber risk distributes according to lateral connectivity, which is why Kentucky, West Virginia, and Pennsylvania consistently experience the highest loads. The multiple alternate paths soften the instability but also ensure that these central nodes absorb most of the disturbance traffic. In the flow model, randomness does not matter because the physical network behaves like a directional chain. Any upstream disruption automatically travels downstream toward the high-influence states with large eigenvector values such as Pennsylvania (0.439) and New Jersey (0.439). This is why the flow model shows $q < 1$ at all vulnerability levels: even unplanned failures accumulate along fixed hydrological routes. The MBRA results therefore confirm that random cyber failures spread widely before concentrating, while random physical failures collapse quickly into the same downstream bottlenecks. These patterns form the baseline for understanding why coordinated physical or cyber-physical attacks produce even more severe effects in the next section.

8. Targeted Attack Analysis

8.1 Targeted Attack on the Highest-Degree Node

Targeted attacks focus on nodes whose structural influence allows failures to spread broadly through the network. Within the ten-state model, Pennsylvania holds the highest combined degree centrality (0.444), strong betweenness in both cascade (0.306) and flow (0.569) models, and significant eigenvector influence. Its removal therefore represents a worst-case scenario for both cyber and physical disruption. Because many shortest paths and upstream-to-downstream chains run through Pennsylvania, a targeted failure at this node produces effects that neither random attacks nor lower-degree failures can replicate. The targeted removal of Pennsylvania therefore serves as the basis for analysing catastrophic fragmentation in both network interpretations.

8.2 Targeted Cyber Attack (Cascade Interpretation)

In the cascade model, a cyber-attack on Pennsylvania behaves like a compromise of a regional operations hub or a supervisory control environment that coordinates cross-state administrative or SCADA-linked activities. Once Pennsylvania is removed, the network loses one of its key lateral bridges, and the load of cyber interaction shifts toward Kentucky and West Virginia. Kentucky's betweenness of 0.639 shows that it becomes the new principal pathway for cross-cluster communication, while West Virginia, with a betweenness of 0.556, inherits a similar role. The removal of Pennsylvania creates an uneven redistribution of cyber pathways that increases stress on nodes originally not designed to absorb this level of interconnection. Cyber intrusion attempts or malware propagation now reach deeper into the interior states because the alternative paths created by the undirected structure must route through these remaining hubs. Although the cascade network does not collapse immediately due to its multiple lateral links, the system becomes structurally imbalanced, and failure propagation becomes faster and more concentrated along the remaining high-betweenness corridors.

8.3 Targeted Cyber Attack (Flow Interpretation)

The flow interpretation produces a far more severe outcome because physical movement follows strict upstream-to-downstream directionality. A targeted physical disruption at Pennsylvania whether contamination, sabotage of a main, or pump-station failure removes the dominant upstream driver for multiple states. West Virginia immediately loses its primary inflow source, which isolates downstream states such as Virginia, Maryland, and New Jersey. Because the flow network has 0% link redundancy, no alternate routing exists to compensate for Pennsylvania's loss. The betweenness value of 0.569 for Pennsylvania reflects this structural chokepoint: once it is removed, the downstream chain collapses into multiple disconnected segments. Kentucky also experiences indirect instability because its inflow-outflow balance changes sharply when regional supply routes upstream of West Virginia degrade. These effects mirror real contamination events like Dunbar, where the absence of hydraulic alternatives forced disruptions to propagate downstream without resistance.

8.4 Implications for Network Fragmentation and System Stability

The contrast between cascade and flow behaviour under targeted attack highlights the different mechanisms of cyber and physical propagation. In the cascade network, redundancy and nondirectional links delay full fragmentation, but the redistribution of cyber pathways forces Kentucky and West Virginia into structurally overloaded positions. Their centrality values confirm that they become the remaining "bridges," and failure pressure concentrates on them. By contrast, in the flow network, fragmentation is immediate. Once Pennsylvania fails, the downstream corridor splits into isolated clusters because no bypass exists for physical movement. The rapid transition into instability is consistent with the negative critical vulnerability (-1.44) derived from MBRA and with the earlier node-robustness findings showing that several states function as blocking nodes. These targeted-attack outcomes demonstrate that the network is cyber-resilient but physically brittle. Cyber-attacks redistribute risk, while physical attacks sever essential flow links and disrupt service across multiple states. This asymmetry underscores the need for DHS to prioritize upstream physical assets while simultaneously strengthening cyber protections on nodes with high informational betweenness.

9. Threat, Vulnerability, Consequence, Prevention, and Response Inputs

9.1 Threat Input

The threat value in the Model Based Risk Analysis framework was assigned a fixed value of 100 for all nodes. This constant serves as a neutral scaling factor and follows the structure used in MBRA, where threat is kept uniform so that all variation in risk arises from network structure, vulnerability, and consequence rather than from probabilistic assumptions. Setting threat to a constant also ensures that comparisons between the cascade and flow results are driven entirely by differences in topology and consequence, rather than changes in threat likelihood. The value does not represent a real-world probability; instead, it establishes a stable baseline that supports regression fitting, critical vulnerability estimation, and comparison across resilience curves.

9.2 Vulnerability Input

The vulnerability inputs were set at two required levels: $\gamma = 0.20$ and $\gamma = 0.80$. These two points serve as the calibration anchors for the MBRA resilience equation

$$\log(q) = b + k \cdot (\gamma \cdot \rho).$$

The lower value of twenty percent represents mild cyber probing, isolated malware events, phishing exposure, minor operational degradation, or low-intensity physical stress. The eighty percent level represents severe cyber events such as sustained ransomware campaigns, SCADA compromise, multistate intrusion attempts, and major physical disruptions such as upstream contamination, main ruptures, or major pumping-station failures. Using identical vulnerability points for both network types ensures that the differences between cascade and flow resilience curves arise solely from structural and hydrological properties rather than inconsistencies in vulnerability assumptions.

9.3 Consequence Input

The consequence values were grounded in real physical and economic losses using the Dunbar contamination incident in West Virginia. The rupture of the 36-inch main caused an estimated 24.9 million gallons per day reduction in service, which MBRA uses as a physical consequence baseline. This provides a measurable, sector-accurate quantity for modelling physical loss. To convert hydraulic disruption into economic terms, the Dunbar event's approximately 18 million dollars in emergency cost was used as the financial anchor. This allowed a consistent gallons-to-dollars conversion ratio to be applied across the entire network. To distribute consequences across states, the Dunbar baseline was scaled using actual financial and infrastructure data from the American Water 2024 Annual Report. States with larger treatment plants, more extensive main networks, or higher capital depreciation received proportionally higher consequence values, reflecting their larger operational exposure. This ensured that consequence inputs remained tied to real capacity and financial structure rather than arbitrary assumptions, allowing MBRA to compute realistic loss estimates under cyber and physical propagation.

9.4 Prevention Cost Input

Prevention cost values were derived from American Water's 2,856 million dollars in capital expenditures (CAPEX) reported for 2024. CAPEX includes treatment facility upgrades, main replacements, pumping improvements, digital monitoring enhancements, and cybersecurity modernization initiatives. Because CAPEX directly reduces vulnerability by strengthening assets and introducing protective measures, it functions as the prevention-cost component in the MBRA model. The total CAPEX amount was allocated proportionally across states according to their relative share of capital footprint in the annual report. This method produces state-level prevention values that reflect each state's real investment posture and aligns with DHS expectations for linking resilience improvement to capital planning.

9.5 Response Cost Input

Response costs were derived from the operational and maintenance expenses and emergency repair activities described in the 2024 Annual Report. American Water reported approximately 152 million dollars in O&M and incident-related obligations that represent the cost required to restore service, repair damaged mains, rebuild pump systems, clean contaminated intakes, or mobilize emergency field teams. These values capture the typical resources needed after events such as contamination, mechanical failure, or cyber-induced operational disruption. The Dunbar event also demonstrated the scale of emergency response cost associated with a single upstream failure, reinforcing the decision to integrate both typical O&M cost and major event cost into the response-cost term. These combined values provide a realistic and defensible estimate of the economic burden associated with physical or cyber incidents at each node.

9.6 Summary of Input Integration

Together, the fixed threat value, two-level vulnerability settings, Dunbar-calibrated consequence baseline, CAPEX-based prevention cost, and O&M-based response cost create a complete and realistic input structure for MBRA. These values ensure that the resulting risk, resilience, and critical vulnerability estimates directly reflect real-world financial exposure, physical capacity, and operational response behaviour in the water sector. They also provide continuity between the analytical foundations in the earlier sections and the risk-ranking and control-allocation results presented in the next stages of the paper.

10. MBRA Rankings and Risk Reduction

10.1 Node Ranking Across Cascade and Flow Models

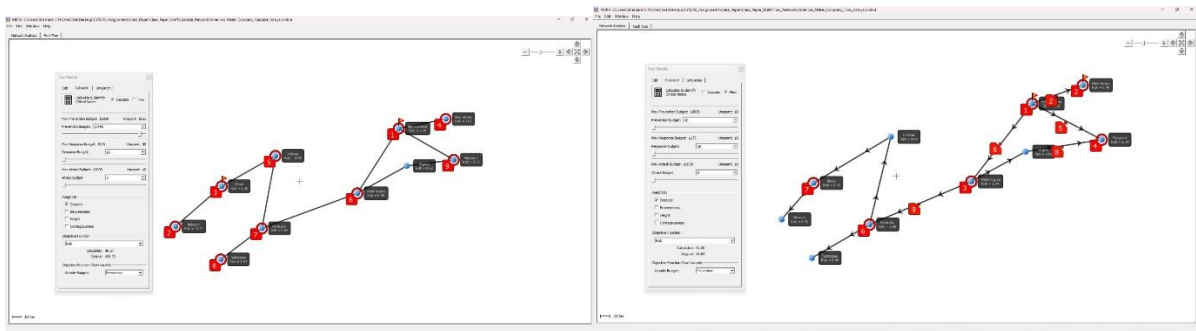


Figure 6: Network Simulations of the Cascade and Flow Models of the American Water Company Network

The MBRA ranking shows consistent structural patterns across both cyber-cascade and physical-flow interpretations, although the reasons for node dominance differ. Pennsylvania emerges as the highest-ranked node in every evaluation because it carries the strongest cyber cross-connections in the cascade model and controls the main upstream-to-downstream pathways in the flow model. This dual importance concentrates consequence propagation around Pennsylvania regardless of attack type or vulnerability level. States such as Illinois and West Virginia follow closely in the cascade ranking because their high betweenness values position them as unavoidable transit points for informational movement. In the flow model, New Jersey rises in importance because it receives the final downstream routing from Pennsylvania and directs it into the Atlantic corridor, giving it a strong influence over physical stability even though its cyber connectivity is low.

10.2 Risk Values Before Prevention Controls

The initial MBRA risk outputs reflect the network's structural fragility before any controls are applied. In the cascade model, the aggregated system risk is approximately 458, which indicates that cyber disruptions can spread widely across the undirected structure when no prevention measures are in place. Pennsylvania and Illinois hold the highest individual risk scores because their centrality positions allow disturbances to reach multiple neighbouring states. In the flow model, the total initial risk is approximately 32. Although the numeric magnitude is lower, the physical consequences are more severe because the directional structure produces immediate downstream collapse. Pennsylvania and New Jersey carry the highest baseline risk because they anchor the upstream and midstream physical pathways that support the entire eastern corridor.

10.3 Risk Values After Prevention Controls

Once the full 2,856 prevention investment is allocated, the MBRA results show a dramatic reduction in system-wide risk. In the cascade model, the total risk falls from approximately 458 to around 40, demonstrating that cyber controls such as access management, network segmentation, and authentication hardening sharply reduce the ability of disturbances to move across lateral connections. In the flow model, the total risk decreases from roughly 32 to about 2, confirming that physical protections such as upstream sensors, valve isolation capability, and treatment-plant reinforcement prevent disruptions from travelling along mandatory hydraulic paths. The far lower final risk values in both models highlight that structural vulnerability can be mitigated even when the underlying network topology remains unchanged.

10.4 Interpretation of Risk Shifts After Controls

The shift in risk patterns after prevention investment reveals how cyber and physical systems respond differently to targeted controls. In the cascade model, risk reduction is broad because cyber controls interrupt multiple informational pathways at once. The remaining risk primarily concentrates around interior nodes such as Kentucky and West Virginia because they continue to serve as bridges between clusters even after prevention reduces vulnerability. In the flow model, the reduction is even sharper because the controls prevent downstream propagation at the earliest possible point. Pennsylvania remains important, but the risk at New Jersey and Maryland decreases substantially because upstream stability prevents downstream inheritance of failures. The overall percentage reduction exceeds ninety percent in both models, showing that strategic investment guided by MBRA rankings can significantly enhance system resilience.

11. Fault Tree Analysis

11.1 Overview of Fault Tree Method and Node Selection

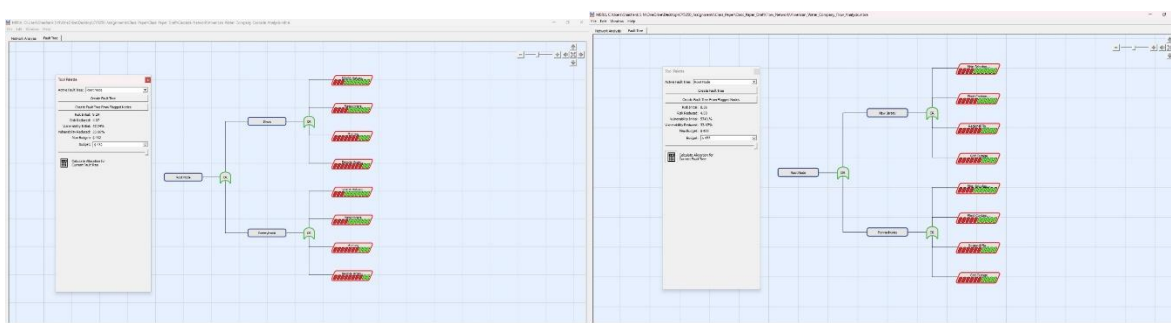


Figure 7: Fault Tree Simulations of the Cascade and Flow Models of the American Water Company Network

The fault tree analysis extends the MBRA findings by identifying which specific cyber and physical threats must be removed to reduce vulnerability at the states that exert the highest structural and consequence influence. Because cyber propagation follows the cascade network, the cyber fault tree focuses on Pennsylvania and Illinois, the two nodes with the highest combined cyber betweenness, eigenvector centrality, and influence over cross-regional digital interactions. Physical propagation follows the directional flow network, so the physical fault tree centers on Pennsylvania and New Jersey, the two states that control most upstream-to-downstream routing and dominate edge-betweenness values in the physical model. To maintain consistency with earlier MBRA settings, the analysis uses the same 40% cyber + 40% physical vulnerability distribution applied in the resilience calibration.

11.2 Cyber Fault Tree Results (Cascade Model)

The cyber fault tree traces how cyber-initiated failures at Pennsylvania and Illinois can emerge from four core threat branches that reflect real vulnerabilities observed in the water sector: SCADA malware injection, ransomware, phishing-based credential compromise, and remote-access exploitation. Both Pennsylvania and Illinois use an OR-gate structure, meaning that the failure of any one branch is sufficient to compromise the node's cyber functionality. This structure aligns with modern water-sector incident reports, where attackers rarely need multiple exploit paths to manipulate pumps, disable alarms, or alter chemical dosing. The cost to eliminate cyber threats was derived from the node-level scaling of American Water's 2024 CAPEX values. Fully eliminating cyber threats in Pennsylvania required \$209.7M, while Illinois required \$222.5M, producing a combined cyber-prevention allocation of \$432M. These investments represent improvements such as SCADA segmentation, privileged-access control, MFA for operators, advanced anomaly detection, and vendor-connection hardening. When the cyber elimination budget is applied, the MBRA software shows a reduction in cyber vulnerability from 57.01% to 33.67%, with total cyber risk falling from 9.24 to 4.07. This reduction is consistent with the earlier cascade-network prevention results, where cyber-oriented investments sharply reduced cross-regional propagation. The results confirm that Pennsylvania and Illinois form the core cyber-influence pair, and eliminating their cyber threat branches has the highest system-wide impact.

11.3 Physical Fault Tree Results (Flow Model)

Physical disruptions follow strict upstream-to-downstream movement, so the physical fault tree focuses on Pennsylvania and New Jersey, the two states that shape most of the downstream hydraulic structure. Each node contains four major physical threat branches: main sabotage, treatment-plant contamination, regional flooding, and upstream electrical grid outage. As in the cyber tree, each branch is modelled through an OR-gate, because the physical failure of any one branch is enough to produce sustained service loss or downstream contamination. The budget calculations mirror the cyber case because the same CAPEX scaling was applied: \$209.7M for Pennsylvania and \$222.5M for New Jersey, again totalling \$432M for full physical threat elimination. These prevention costs correspond to asset hardening (e.g., thicker-wall mains and flood barriers), upstream valve control, distributed water quality monitoring, and physical security upgrades at critical pump stations and treatment plants. After prevention measures are applied, MBRA shows the physical vulnerability dropping from 57.24% to 33.66%, and total physical risk declining from 8.36 to 4.08. This reduction directly reflects the system's dependence on upstream nodes: once Pennsylvania and New Jersey are hardened, directional failure propagation is sharply reduced, and downstream states receive more stable inflow.

11.4 Combined Prevention and Response Control Matrix

The combined effect of cyber and physical threat elimination demonstrates that targeted prevention focused on structurally dominant nodes yields the highest return on investment. Across both fault trees, three nodes consistently emerge as topological and operational priorities: Pennsylvania, Illinois, and New Jersey. These states sit at the intersection of high centrality, high consequence, and high vulnerability, making them the most effective targets for CAPEX-driven risk reduction. Response controls (such as emergency repairs, mobile pumping, rapid substation restoration, or contamination containment) rely on the O&M and Dunbar-derived cost structure already embedded in the MBRA model. When paired with the prevention matrix, they create a two-layer mitigation strategy that reduces both the probability of failure and the severity of downstream impacts. Together, the two fault trees confirm that focusing prevention and response resources on PA, IL, and NJ provides the best system-wide reduction in vulnerability and supports the overall DHS objective of stabilizing both cyber and physical lifeline functions.

12. Budget for Resilience Improvement

The prevention budget derived from American Water's 2024 capital expenditure total of 2,856 million dollars provides a strong foundation for reducing system vulnerability, but it is not sufficient to eliminate all major cyber and physical threats in the regional network. The fault tree analysis showed that removing the four major cyber threats at Pennsylvania and Illinois requires 432 million dollars. Eliminating the four major physical threats at Pennsylvania and New Jersey requires an additional 432 million dollars. Together, these two elimination strategies amount to 864 million dollars, which already exceeds the targeted prevention portion available within the CAPEX budget. This finding demonstrates that the regional water system contains more high-impact vulnerabilities than the current investment level can fully address.

Because the available prevention allocation cannot fund complete threat removal at all critical nodes, additional resources will be necessary through DHS support or federal and state cost sharing. Prioritization therefore becomes essential. Pennsylvania requires the first allocation because it dominates system behaviour in both cascade and flow models and produces the highest reduction in regional risk when hardened. Illinois should receive the second allocation because it is one of the strongest cyber conduits in the cascade model. New Jersey should follow next because it carries major downstream responsibility in the flow model and stabilizes the physical corridor leading to the Atlantic region. Allocating resources in this order produces the greatest resilience improvement per dollar spent and ensures that limited federal and state funds generate measurable reductions in both cyber and physical vulnerability.

13. Cyber Workforce Development and the Role of NICC and NCCIC

A resilient national water system requires more than technical controls because cyber and physical threats continue to evolve faster than many utilities can respond. Workforce capability therefore becomes a central part of national resilience. The water sector continues to face shortages of specialists who can manage supervisory control systems, OT network security, incident response, and cross sector coordination. American Water and other utilities depend heavily on a small group of engineers who maintain both digital and physical assets. This narrow staffing structure increases operational risk because the loss of a small number of personnel can weaken monitoring, delay patching, and reduce the ability to respond to cyber intrusions or physical disruptions. Strengthening the cyber workforce is therefore a core component of long-term stability in both the cascade and flow models. The highest priority roles are SCADA security analysts, OT network engineers, threat detection specialists, and cross domain responders who can manage cyber events that influence physical movement. Federal support through the Cybersecurity and Infrastructure Security Agency plays an important role in filling these gaps. The National Cybersecurity and Communications Integration Center provides national level situational awareness and real time cyber threat analysis. It supplies alerts, indicators of compromise, forensic guidance, and monitoring support that many regional utilities cannot produce on their own. The lateral nature of cyber propagation shown in the cascade model makes this coordination essential because compromises that appear at one utility often spread across shared vendors or cloud managed control systems. The NCCIC feeds these alerts directly to sector owners which gives local operators early warning before a threat reaches operational systems. This external support increases the effectiveness of the prevention strategies developed earlier because it allows internal teams to respond faster when cyber pathways begin to show abnormal activity.

The National Infrastructure Coordinating Center performs a complementary role by linking cyber information with the physical status of lifeline sectors. The flow model shows that upstream failures produce immediate downstream impacts. The NICC helps identify these cross-sector dependencies by coordinating information between water utilities, energy providers, and emergency management agencies. During a physical disruption such as a pump failure or a main rupture the NICC ensures that cyber teams, plant operators, and regional response organizations share a common picture of the event. This prevents fragmented decision making and improves the ability to isolate systems, deploy field teams, or shift operational loads. Together the workforce investment and the federal coordination functions of the NICC and NCCIC create a combined framework that strengthens both cyber and physical resilience. A trained workforce protects the control environment from intrusion. The federal centers provide the shared intelligence and cross sector visibility that allow utilities to respond to cascading or downstream disruptions. For a system shaped by the structural dependencies identified earlier these capabilities form the foundation for a national strategy that can absorb cyber failures, manage physical disruptions, and maintain stability across the entire network.

14. Return on Investment & DHS Recommendation

The return on investment for the prevention and response strategy is evaluated by comparing the reduction in system wide risk achieved through targeted controls against the total investment required to implement those controls. The MBRA outputs provide a clear numerical basis for this comparison. In the cascade network the total system risk declined from approximately 458 before prevention to about 40 after prevention. In the flow network the total system risk declined from approximately 32 to about 2 after prevention. These reductions demonstrate that relatively concentrated actions at a few structural hubs can produce meaningful system wide improvements even though the underlying topology was not modified. The improvement also reflects the fact that cyber controls interrupt many lateral propagation routes at once and physical controls restrict forward movement along the fixed downstream chain.

The return on investment can be expressed by using a simple risk reduction formula:

$$ROI = \frac{Risk_{before} - Risk_{after}}{Investment}$$

This formula can be applied in both economic and operational terms. When expressed in dollars the cyber related risk reduction of roughly four hundred and eighteen units achieved through targeted investments represents a strong return relative to the 2,856-million-dollar prevention budget. When expressed in gallons the Dunbar baseline of 24.9 million gallons per day provides a meaningful operational measure. Reducing system risk by ninety percent prevents the equivalent of several Dunbar scale losses when scaled across all ten nodes. The combined cyber and physical risk reductions indicate that each dollar invested in upstream protection prevents many millions of gallons of lost service in downstream states. This is significant because the economic cost of service loss is non-linear and escalates rapidly when multiple regions are affected simultaneously.

The physical return on investment is particularly strong because upstream controls at Pennsylvania and West Virginia protect every downstream user. A reinforced treatment plant or an upgraded pump station in an upstream state prevents the loss of millions of gallons of daily supply to Maryland, Virginia, and New Jersey. The cyber return on investment is also measurable because controls such as authentication, logging, and segmentation reduce the probability of large scale administrative or SCADA compromise. When cyber controls limit lateral movement, they delay the onset of instability in the cascade model and support a more coordinated operational response to physical events.

Based on these results the Department of Homeland Security should prioritise prevention spending in the states that have the greatest structural influence over both cyber and physical propagation. Pennsylvania must remain the primary investment location because it dominates both cascade and flow networks. Illinois should receive dedicated cyber investment because its centrality and eigenvector structure make it a major channel for lateral cyber movement. New Jersey requires targeted physical investment because it forms the final major downstream transfer point in the regional corridor. DHS should consider a combined federal–state funding approach that directs resources to these structural hubs while maintaining alignment with NICC and NCCIC guidance for integrated cyber and physical risk monitoring. A coordinated approach will ensure that prevention spending yields the maximum reduction in gallons lost, service outages, and economic disruption across the entire regional water network.

15. Conclusion

This study applied complex network analysis and Model Based Risk Analysis to examine how cyber and physical disruptions propagate through a ten-state representation of the American Water system. By grounding the analysis in operational data from the Dunbar contamination event and in the capital and operational investments reported in the American Water 2024 Annual Report, the model produced a defensible assessment of system behaviour. The results show that the cyber based cascade network retains stability across most vulnerability levels, while the physical flow network becomes unstable even when stress is low. The difference arises from the underlying structure. Cyber pathways contain multiple lateral routes that distribute load across several states. Physical pathways follow strict upstream to downstream movement where a failure at a single upstream location transfers its full impact to downstream regions without any possibility of rerouting.

Across all analyses Pennsylvania emerged as the dominant node for both cyber and physical propagation. Illinois and New Jersey provided added influence in their respective models, and Kentucky and West Virginia acted as internal conduits that connect multiple regions and concentrate disturbance movement. The random and targeted attack simulations confirmed that failures at these central locations create the largest regional consequences and rapid shifts in downstream stability. The fault tree evaluations and the prevention cost calculations also showed that the most effective investments target these high influence nodes because removing even a subset of threats at these locations produces measurable reductions in overall system vulnerability.

Although the available prevention budget based on the 2024 capital expenditure can significantly reduce risk it cannot eliminate all major threats. The remaining vulnerability underscores the need for sustained federal support and coordinated planning between DHS, NICC, and NCCIC. Strengthening cyber workforce capacity, improving cyber physical situational awareness, and expanding upstream monitoring and rapid isolation capabilities create measurable returns in both gallons protected and economic value preserved. The findings show that resilience improvement is achievable through targeted investment and informed prioritization. The overall conclusion is that DHS should continue to advance integrated cyber and physical protection strategies focused on the small set of nodes that anchor system wide stability to enhance national water security and safeguard the communities that depend on it.

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Appendix:

1. Python Code

```
import numpy as np
import numpy.linalg as npla
import networkx as nx

# Node order:
# 1: Illinois
# 2: Indiana
# 3: Kentucky
# 4: Maryland
# 5: Missouri
# 6: New Jersey
# 7: Pennsylvania
# 8: Tennessee
# 9: West Virginia
# 10: Virginia

C = np.array([
    [0,1,0,0,1,0,0,0,0,0],
    [1,0,1,0,0,0,0,0,0,0],
    [0,1,0,0,0,0,0,1,1,0],
    [0,0,0,0,0,0,1,0,0,1],
    [1,0,0,0,0,0,0,0,0,0],
    [0,0,0,0,0,0,1,0,0,0],
    [0,0,0,1,0,1,0,0,0,1],
    [0,0,1,0,0,0,0,0,0,0],
    [0,0,1,0,0,0,1,0,0,1],
    [0,0,0,1,0,0,0,0,1,0]
])

print("CASCADE NETWORK\n")
print("Adjacency Matrix (C):")
print(C)
print()
```

```

# Node degrees (undirected)
degrees = np.sum(C, axis=1)
for i, deg in enumerate(degrees, start=1):
    print(f'Degree of Node {i} = {deg}')
NetworkDegree = np.max(degrees)
print("\nDegree of Network =", NetworkDegree)
# Eigenvalues + Spectral radius
eigenvalues, eigenvectors = npla.eig(C)
print("\nEigenvalues of C =\n", eigenvalues)
spectral_radius = np.max(eigenvalues)
print("\nSpectral Radius =", spectral_radius)
print("\nEigenvectors of C =\n", eigenvectors)
# Build undirected graph
G = nx.from_numpy_array(C)
# Degree Centrality
DC = nx.degree_centrality(G)
print("\nDegree Centrality (Undirected) =")
print(DC)
# Betweenness Centrality
BC = nx.betweenness_centrality(G, normalized=True)
print("\nBetweenness Centrality (Undirected) =")
print(BC)
# Eigenvector Centrality
EC = nx.eigenvector_centrality_numpy(G)
print("\nEigenvector Centrality (Undirected) =")
print(EC)
print("\n\nDIRECTED FLOW NETWORK ANALYSIS\n")
# Convert undirected adjacency into directed (same edges but bidirectional where C=1)
CF = C.copy()
GF = nx.from_numpy_array(CF, create_using=nx.DiGraph)
# In-Degree & Out-Degree
print("\nDirected Node Degrees:")
for i in range(10):
    print(f'Node {i+1}: In-Degree = {GF.in_degree(i)}, Out-Degree = {GF.out_degree(i)}')

```

```

# Directed Degree Centrality
DC_in = nx.in_degree_centrality(GF)
DC_out = nx.out_degree_centrality(GF)
print("\nIn-Degree Centrality (Directed):")
print(DC_in)
print("\nOut-Degree Centrality (Directed):")
print(DC_out)

# Directed Betweenness Centrality (most important for flow)
BC_directed = nx.betweenness_centrality(GF, normalized=True)
print("\nBetweenness Centrality (Directed) =")
print(BC_directed)

# Directed Eigenvector Centrality
EC_directed = nx.eigenvector_centrality_numpy(GF)
print("\nEigenvector Centrality (Directed) =")
print(EC_directed)

# Link Betweenness (Directed)
edge_bet = nx.edge_betweenness_centrality(GF, normalized=True)
print("\nLink (Edge) Betweenness Centrality (Directed):")
for edge, val in edge_bet.items():
    print(f'Edge {edge} = {val}')

```

2. Python Output

Shashank Shivakumar Niveditha: Class Paper Draft Submission

CASCADE NETWORK

Adjacency Matrix (C):

```

[[0 1 0 0 1 0 0 0 0 0]
 [1 0 1 0 0 0 0 0 0 0]
 [0 1 0 0 0 0 0 1 1 0]
 [0 0 0 0 0 0 1 0 0 1]
 [1 0 0 0 0 0 0 0 0 0]
 [0 0 0 0 0 0 1 0 0 0]
 [0 0 0 1 0 1 0 0 0 1]
 [0 0 1 0 0 0 0 0 0 0]
 [0 0 1 0 0 0 1 0 0 1]
 [0 0 0 1 0 0 0 0 1 0]]

```

Degree of Node 1 = 2

Degree of Node 2 = 2

Degree of Node 3 = 3

Degree of Node 4 = 2

Degree of Node 5 = 1

Degree of Node 6 = 1

Degree of Node 7 = 3

Degree of Node 8 = 1

Degree of Node 9 = 3

Degree of Node 10 = 2

Degree of Network = 3

Eigenvalues of C =

[2.27295648e+00+0.j 1.75074248e+00+0.j
1.12469091e+00+0.j 4.88722194e-01+0.j
-1.93378497e+00+0.j -1.00000000e+00+0.j
-1.35166354e+00+0.25369507j -1.35166354e+00-0.25369507j
-1.69465381e-18+0.j -1.86896739e-17+0.j]

Spectral Radius = (2.2729564807812657+0j)

Eigenvectors of C =

[[1.31510416e-01+0.j 4.39584690e-01+0.j
5.43989787e-01+0.j 1.81388119e-02+0.j
3.38042116e-01+0.j -4.08248290e-01+0.j
-3.32435382e-01-0.03515726j -3.32435382e-01+0.03515726j
6.64795845e-17+0.j 1.24659243e-17+0.j]
[2.41058704e-01+0.j 5.18514866e-01+0.j
1.28140998e-01+0.j -2.82499302e-02+0.j
-4.78892230e-01+0.j -5.29162906e-16+0.j
2.25399736e-01-0.10653257j 2.25399736e-01+0.10653257j
-2.32075069e-01+0.j 5.75221480e-01+0.j]
[4.16405527e-01+0.j 4.68201310e-01+0.j
-3.99870772e-01+0.j -3.19451798e-02+0.j
5.88032483e-01+0.j 4.08248290e-01+0.j
5.47975665e-02+0.23633626j 5.47975665e-02-0.23633626j
-4.04890218e-17+0.j -2.93713849e-17+0.j]

[3.39013026e-01+0.j -2.33675230e-01+0.j
 1.33204722e-01+0.j 1.43367013e-01+0.j
 -5.00687561e-02+0.j 4.08248290e-01+0.j
 -3.44503878e-01+0.15348477j -3.44503878e-01-0.15348477j
 -5.19507715e-01+0.j -6.26185346e-03+0.j]
 [5.78587477e-02+0.j 2.51084723e-01+0.j
 4.83679368e-01+0.j 3.71147702e-02+0.j
 -1.74808534e-01+0.j 4.08248290e-01+0.j
 2.32860273e-01+0.06971614j 2.32860273e-01-0.06971614j
 2.32075069e-01+0.j -5.75221480e-01+0.j]
 [1.72313297e-01+0.j -1.68450484e-01+0.j
 2.03666291e-01+0.j -7.83448551e-01+0.j
 5.80255291e-02+0.j -4.12334732e-16+0.j
 2.37337620e-02+0.22259966j 2.37337620e-02-0.22259966j
 5.19507715e-01+0.j 6.26185346e-03+0.j]
 [3.91660624e-01+0.j -2.94913418e-01+0.j
 2.29061625e-01+0.j -3.82888694e-01+0.j
 -1.12208896e-01+0.j 4.53568205e-16+0.j
 -8.85524977e-02-0.2948587j -8.85524977e-02+0.2948587j
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 [1.83199956e-01+0.j 2.67430142e-01+0.j
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 [5.22212982e-01+0.j 3.37549124e-02+0.j
 -2.22333648e-01+0.j 7.80023123e-02+0.j
 -3.54152453e-01+0.j 7.37048333e-16+0.j
 -3.51964416e-01-0.02276394j -3.51964416e-01+0.02276394j
 5.19507715e-01+0.j 6.26185346e-03+0.j]
 [3.78901231e-01+0.j -1.14191733e-01+0.j
 -7.92474858e-02+0.j 4.52955335e-01+0.j
 2.09031104e-01+0.j -4.08248290e-01+0.j
 5.15267500e-01+0.j 5.15267500e-01-0.j
 4.93021427e-16+0.j -1.36095976e-16+0.j]]

Degree Centrality (Undirected) =

{0: 0.2222222222222222, 1: 0.2222222222222222, 2: 0.3333333333333333, 3: 0.2222222222222222, 4: 0.1111111111111111, 5: 0.1111111111111111, 6: 0.4444444444444444, 7: 0.1111111111111111, 8: 0.3333333333333333, 9: 0.3333333333333333}

Betweenness Centrality (Undirected) =

{0: 0.2222222222222222, 1: 0.3888888888888888, 2: 0.6388888888888888, 3: 0.0, 4: 0.0, 5: 0.0, 6: 0.3055555555555555, 7: 0.0, 8: 0.5555555555555556, 9: 0.0833333333333333}

Eigenvector Centrality (Undirected) =

{0: 0.039653519057066655, 1: 0.09611057413057503, 2: 0.22782931469674836, 3: 0.3742303073880032, 4: 0.014248101193589376, 5: 0.1964442539750863, 6: 0.5467188829454138, 7: 0.08186247293698572, 8: 0.45609278032498884, 9: 0.49479173128428844}

DIRECTED FLOW NETWORK ANALYSIS

Directed Node Degrees:

Node 1: In-Degree = 2, Out-Degree = 2

Node 2: In-Degree = 2, Out-Degree = 2

Node 3: In-Degree = 3, Out-Degree = 3

Node 4: In-Degree = 2, Out-Degree = 2

Node 5: In-Degree = 1, Out-Degree = 1

Node 6: In-Degree = 1, Out-Degree = 1

Node 7: In-Degree = 3, Out-Degree = 3

Node 8: In-Degree = 1, Out-Degree = 1

Node 9: In-Degree = 2, Out-Degree = 3

Node 10: In-Degree = 3, Out-Degree = 2

In-Degree Centrality (Directed):

{0: 0.2222222222222222, 1: 0.2222222222222222, 2: 0.3333333333333333, 3: 0.2222222222222222, 4: 0.1111111111111111, 5: 0.1111111111111111, 6: 0.3333333333333333, 7: 0.1111111111111111, 8: 0.2222222222222222, 9: 0.3333333333333333}

Out-Degree Centrality (Directed):

{0: 0.2222222222222222, 1: 0.2222222222222222, 2: 0.3333333333333333, 3: 0.2222222222222222, 4: 0.1111111111111111, 5: 0.1111111111111111, 6: 0.3333333333333333, 7: 0.1111111111111111, 8: 0.3333333333333333, 9: 0.2222222222222222}

Betweenness Centrality (Directed) =

{0: 0.2222222222222222, 1: 0.3888888888888888, 2: 0.6388888888888888, 3: 0.0138888888888888, 4: 0.0, 5: 0.0, 6: 0.2638888888888889, 7: 0.0, 8: 0.5694444444444444, 9: 0.2916666666666666}

Eigenvector Centrality (Directed) =

{0: 0.09342021078062325, 1: 0.1712393258601463, 2: 0.2957993246978127, 3: 0.4340378010657823, 4: 0.04110074766962229, 5: 0.19321529122436104, 6: 0.4391699483744528, 7: 0.13013857819052396, 8: 0.37096108803194533, 9: 0.5473790844620694}

Link (Edge) Betweenness Centrality (Directed):

Edge (0, 1) = 0.1777777777777778
Edge (0, 4) = 0.1
Edge (1, 0) = 0.1777777777777778
Edge (1, 2) = 0.2333333333333334
Edge (2, 1) = 0.2333333333333334
Edge (2, 7) = 0.1
Edge (2, 8) = 0.2777777777777778
Edge (3, 6) = 0.0333333333333333
Edge (3, 9) = 0.0777777777777778
Edge (4, 0) = 0.1
Edge (5, 6) = 0.1
Edge (6, 3) = 0.0555555555555556
Edge (6, 5) = 0.1
Edge (6, 9) = 0.1555555555555556
Edge (7, 2) = 0.1
Edge (8, 2) = 0.2777777777777778
Edge (8, 6) = 0.1777777777777778
Edge (8, 9) = 0.1
Edge (9, 3) = 0.0555555555555556
Edge (9, 8) = 0.2777777777777778

3. Derived Data Table

Name	Consequence(MGD)	Consequence (\$M)	Prevention Cost (\$M)	Response Cost (\$M)
Illinois	145.2	105.0	556.3	29.6
Indiana	66.3	48.0	254.1	13.5
Kentucky	23.5	17.0	90.0	4.8
Maryland	24.3	17.6	93.0	5.0
Missouri	170.1	123.0	651.6	34.7
New Jersey	110.6	79.9	423.6	22.5
Pennsylvania	136.9	98.9	524.3	27.9
Tennessee	26.8	19.3	102.5	5.5
West Virginia	24.9	18.0	95.4	5.1
Virginia	17.0	12.3	65.2	3.5
PA-NJ	11.1	8.0	42.4	2.3
PA-MD	2.4	1.8	9.3	13.8
PA-WV	2.5	1.8	9.5	0.5
WV-VA	1.7	1.2	6.5	0.3
VA-MD	1.7	1.2	6.5	0.3
WV-KY	2.4	1.7	9.0	0.5
KY-IN	2.4	1.7	9.0	0.5
IN-IL	6.6	4.8	25.4	1.4
IL-MO	14.5	10.5	55.6	3.0
KY-TN	2.4	1.7	9.0	0.5

4. Derived Fault Tree Data Table

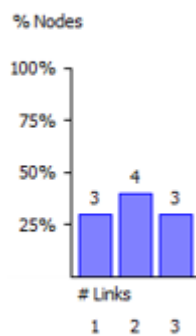
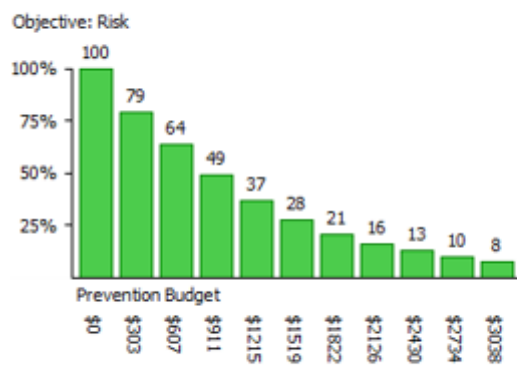
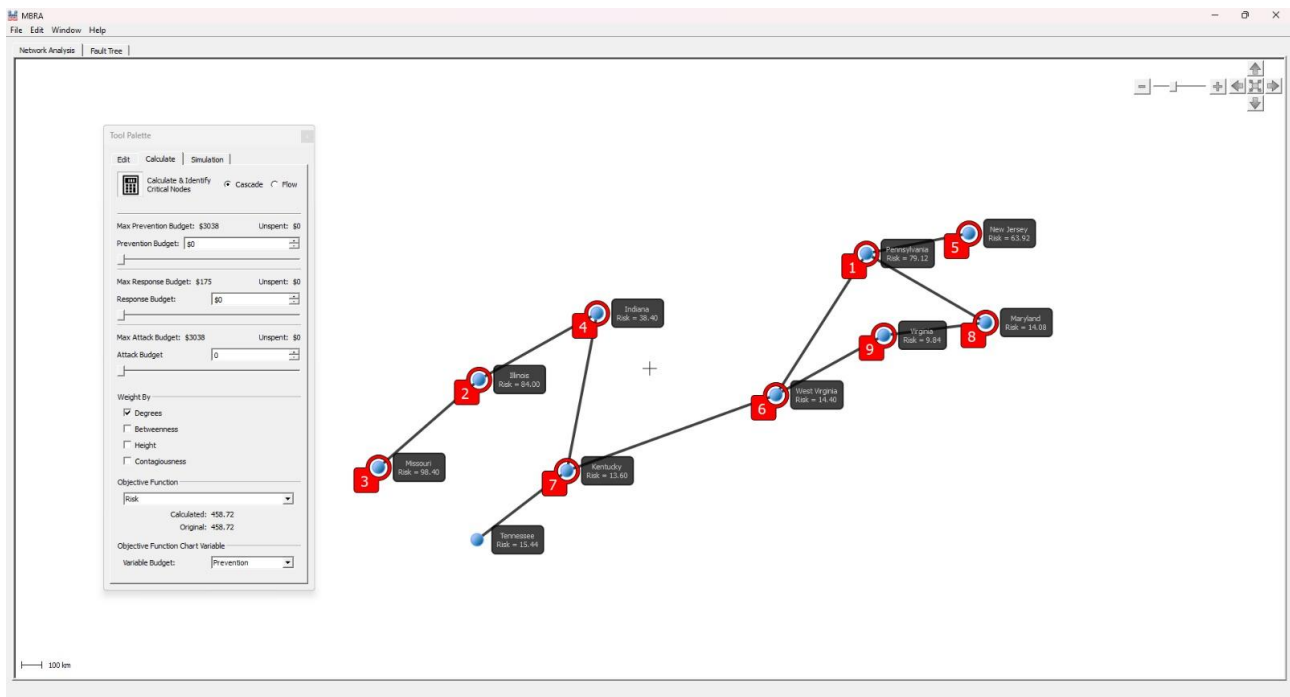
Node	Threat Name	Vulnerability (%)	Elimination Cost (\$M)	Consequence (\$M)
Pennsylvania	SCADA Malware	15.00%	78.6	14.8
	Ransomware	12.00%	62.9	11.9
	Phishing	7.00%	36.7	6.9
	Remote Access	6.00%	31.5	5.9
Total:		40.00%	209.7	39.6
Illinois	SCADA Malware	15.00%	83.4	15.7
	Ransomware	12.00%	66.8	12.6
	Phishing	7.00%	38.9	7.3
	Remote Access	6.00%	33.4	6.3
Total:		40.00%	222.5	42.0

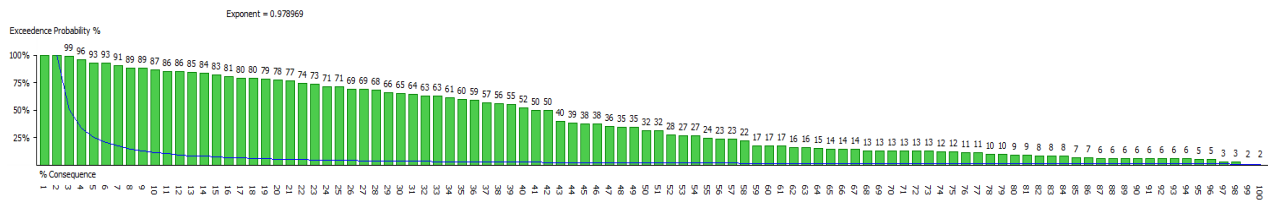
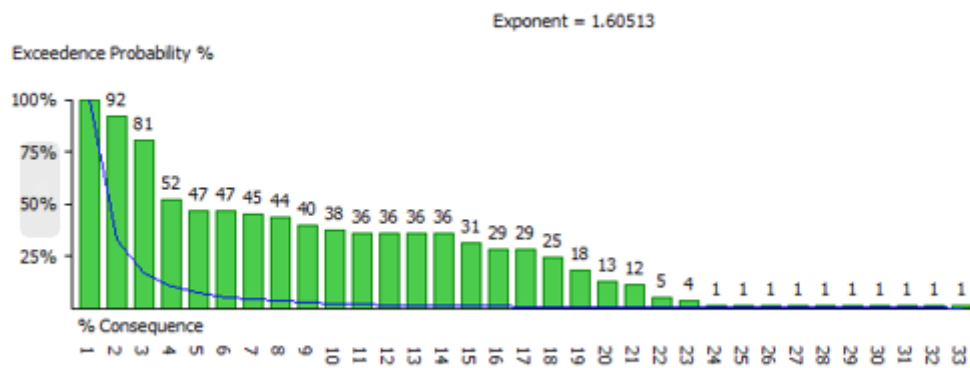
Cascade Network - Cyber Threats (Coordinated and Random)

Node	Threat Name	Vulnerability (%)	Elimination Cost (\$M)	Consequence (\$M)
Pennsylvania	Main Sabotage	12.00%	62.9	11.9
	Plant Contamination	11.00%	57.7	10.9
	Regional Flooding	9.00%	47.2	8.9
	Grid Outage	8.00%	41.9	7.9
Total:		40.00%	209.7	39.6
New Jersey	Main Sabotage	12.00%	66.8	12.6
	Plant Contamination	11.00%	61.2	11.5
	Regional Flooding	9.00%	50.1	9.4
	Grid Outage	8.00%	44.5	8.4
Total:		40.00%	222.5	42.0

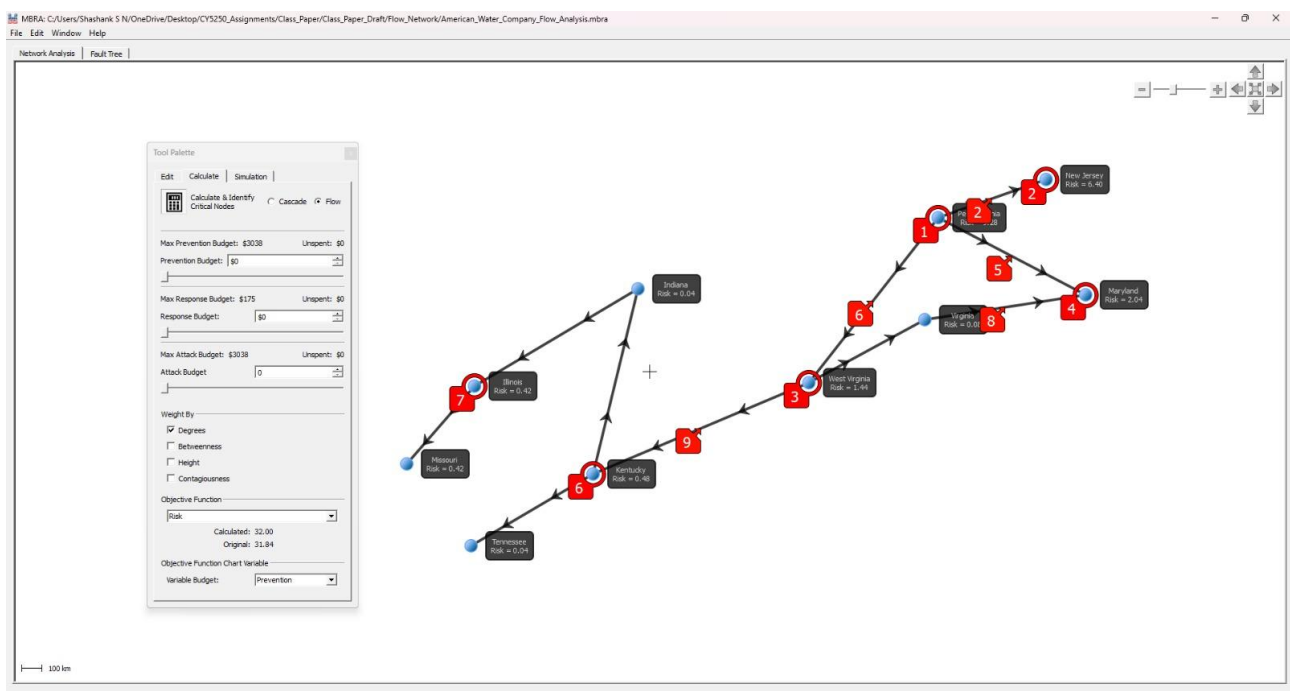
Flow Network - Physical Threats (Coordinated and Random)

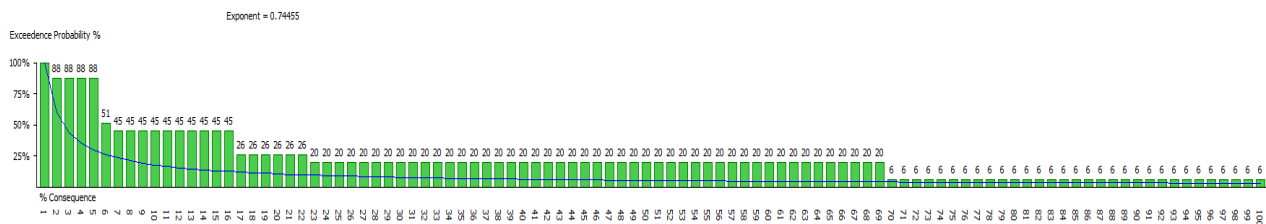
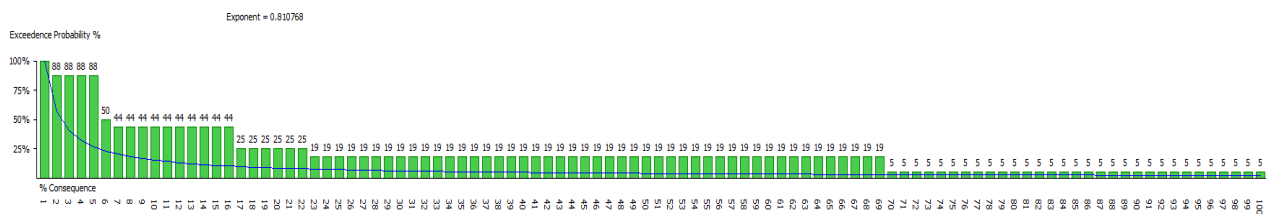
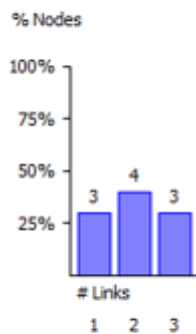
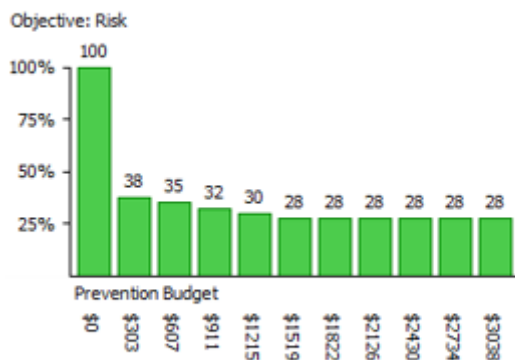
5. Cascade Network Graphs





6. Flow Network Graphs





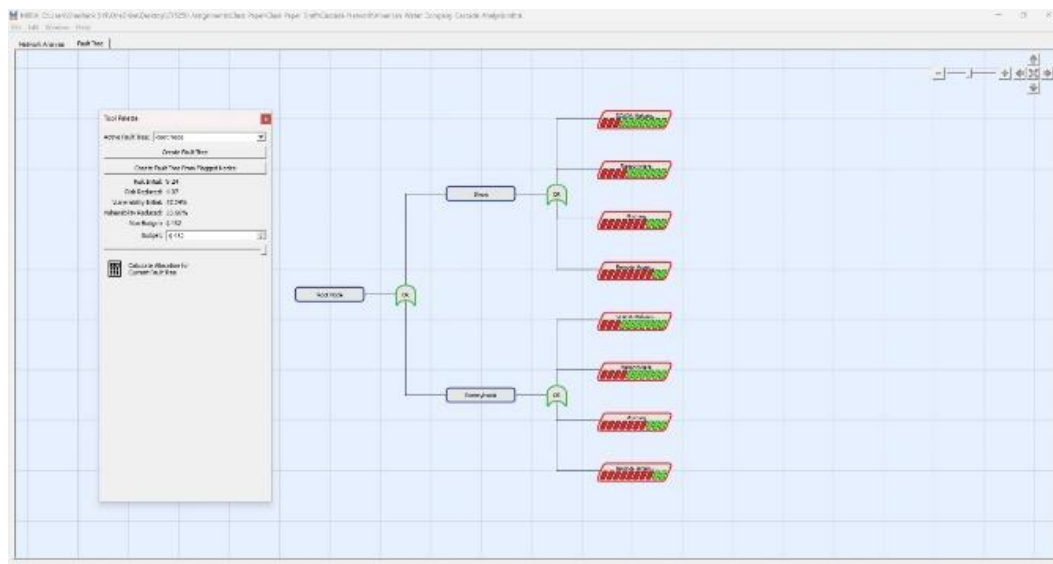
6. Cascade Network Analysis Table

Name	Threat	Vulnerabi	Conseque	Preventio	Response	Risk Initia	Risk Redu	Flow Cons	Preventio	Response	Attack All	Vulnerabi	Conseque	Calculate	Weight	Weight x C	Degrees	Rank
Illinois	100	80	105	556.3	29.6	84	5.25		556.3	0	0	5	105	100	0.667	3.5	2	3
Indiana	100	80	48	254.1	13.5	38.4	2.4		254.1	0	0	5	48	100	0.667	1.6	2	5
Kentucky	100	80	17	90	4.8	13.6	0.85		90	0	0	5	17	100	1	0.85	3	7
Maryland	100	80	17.6	93	5	14.08	0.88		93	0	0	5	17.6	100	0.667	0.587	2	9
Missouri	100	80	123	651.6	34.7	98.4	11.71		500.32	0	0	9.52	123	100	0.333	3.902	1	2
New Jerse	100	80	79.9	423.6	22.5	63.92	7.61		325.14	0	0	9.52	79.9	100	0.333	2.537	1	4
Pennsylv	100	80	98.9	524.3	27.9	79.12	4.95		524.3	0	0	5	98.9	100	1	4.945	3	1
Tennessee	100	80	19.3	102.5	5.5	15.44	1.84		78.61	0	0	9.54	19.3	100	0.333	0.614	1	8
West Virgi	100	80	18	95.4	5.1	14.4	0.9		95.4	0	0	5	18	100	1	0.9	3	6
Virginia	100	80	12.3	65.2	3.5	9.84	0.62		65.2	0	0	5	12.3	100	0.667	0.41	2	10
PA-NJ	100	80	8	42.4	2.3	6.4	0.76		32.55	0	0	9.52	8	100	0.333	0.254	1	12
PA-MD	100	80	1.8	9.3	13.8	1.44	0.17		7.22	0	0	9.28	1.8	100	0.333	0.056	1	15
PA-WV	100	80	1.8	9.5	0.5	1.44	0.17		7.31	0	0	9.48	1.8	100	0.333	0.057	1	14
WV-VA	100	80	1.2	6.5	0.3	0.96	0.12		4.94	0	0	9.73	1.2	100	0.333	0.039	1	17
VA-MD	100	80	1.2	6.5	0.3	0.96	0.12		4.94	0	0	9.73	1.2	100	0.333	0.039	1	17
WV-KY	100	80	1.7	9	0.5	1.36	0.16		6.91	0	0	9.51	1.7	100	0.333	0.054	1	16
KY-IN	100	80	1.7	9	0.5	1.36	0.16		6.91	0	0	9.51	1.7	100	0.333	0.054	1	16
IN-IL	100	80	4.8	25.4	1.4	3.84	0.46		19.51	0	0	9.51	4.8	100	0.333	0.152	1	13
IL-MO	100	80	10.5	55.6	3	8.4	1		42.7	0	0	9.51	10.5	100	0.333	0.333	1	11
KY-TN	100	80	1.7	9	0.5	1.36	0.16		6.91	0	0	9.51	1.7	100	0.333	0.054	1	16

7. Flow Network Analysis Table

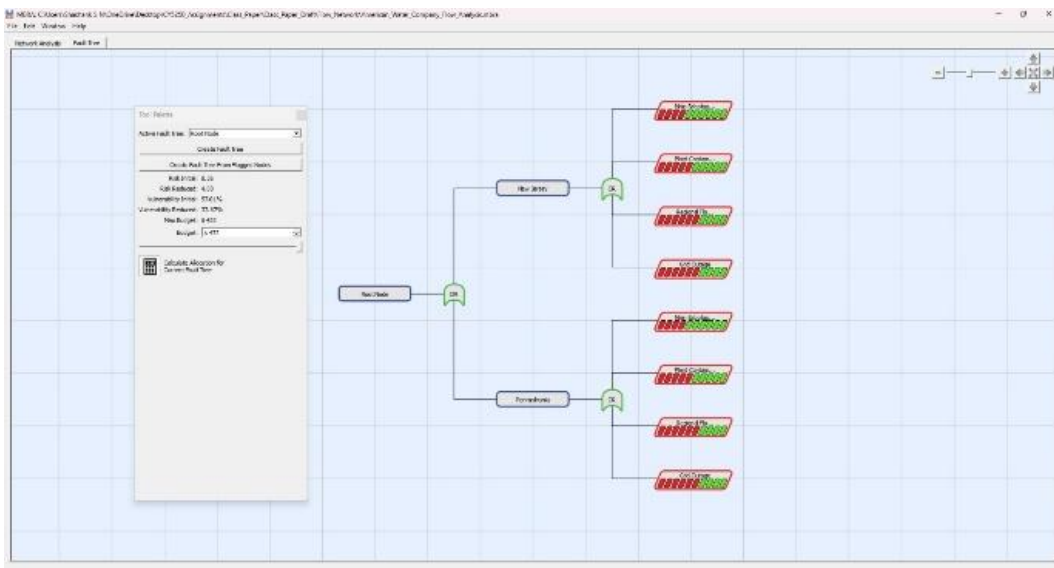
Name	Threat	Vulnerabi	Conseque	Preventio	Response	Risk Initia	Risk Redu	Flow Cons	Preventio	Response	Attack All	Vulnerabi	Conseque	Calculate	Weight	Weight x C	Degrees	Rank
Illinois	100	80	105	556.3	29.6	0.42	0.07	0.53	354.45	0	0	13.67	0.53	100	0.667	0.048	2	6
Indiana	100	80	48	254.1	13.5	0	0.04	0	0	0	0	80	0.05	100	0.667	0.027	2	9
Kentucky	100	80	17	90	4.8	0.48	0.03	0.6	90	0	0	5	0.6	100	1	0.03	3	8
Maryland	100	80	17.6	93	5	2.04	0.13	2.54	93	0	0	5	2.54	100	0.667	0.085	2	5
Missouri	100	80	123	651.6	34.7	0.42	0.42	0.53	0	0	0	80	0.53	100	0.333	0.141	1	2
New Jerse	100	80	79.9	423.6	22.5	6.4	0.4	8	423.6	0	0	5	8	100	0.333	0.133	1	3
Pennsylv	100	80	98.9	524.3	27.9	9.28	0.58	11.6	524.3	0	0	5	11.6	100	1	0.58	3	1
Tennessee	100	80	19.3	102.5	5.5	0	0.04	0	0	0	0	80	0.05	100	0.333	0.013	1	10
West Virgi	100	80	18	95.4	5.1	1.44	0.09	1.8	95.4	0	0	5	1.8	100	1	0.09	3	4
Virginia	100	80	12.3	65.2	3.5	0.08	0.01	0.1	52.85	0	0	8.45	0.1	100	0.667	0.006	2	15
PA-NJ	100	80	8	42.4	2.3	6.4	0.4	8	42.4	0	0	5	8	100	0.333	0.133	1	3
PA-MD	100	80	1.8	9.3	13.8	1.44	0.09	1.8	9.3	0	0	5	1.8	100	0.333	0.03	1	7
PA-WV	100	80	1.8	9.5	0.5	1.44	0.09	1.8	9.5	0	0	5	1.8	100	0.333	0.03	1	8
WV-VA	100	80	1.2	6.5	0.3	0.08	0	0.1	6.5	0	0	5	0.1	100	0.333	0.002	1	16
VA-MD	100	80	1.2	6.5	0.3	0.6	0.04	0.74	6.5	0	0	5	0.74	100	0.333	0.012	1	11
WV-KY	100	80	1.7	9	0.5	0.48	0.03	0.6	9	0	0	5	0.6	100	0.333	0.01	1	12
KY-IN	100	80	1.7	9	0.5	0	0.04	0	0	0	0	80	0.05	100	0.333	0.013	1	10
IN-IL	100	80	4.8	25.4	1.4	0.42	0.03	0.53	25.4	0	0	5	0.53	100	0.333	0.009	1	13
IL-MO	100	80	10.5	55.6	3	0.42	0.03	0.53	55.6	0	0	5	0.53	100	0.333	0.009	1	14
KY-TN	100	80	1.7	9	0.5	0	0.04	0	0	0	0	80	0.05	100	0.333	0.013	1	10

8. Cascade Fault Tree Analysis and Table



Name	Threat	Vulnerabi	Eliminatio	Conseque	Risk Initia	Allocation	Vulnerabi	Risk Redu	Path
SCADA Ma	100	80	524.3	98.9	79.12		80	79.12	Pennsylvania.SCADA Malware
Coordinat	100	80	524.3	98.9	79.12		80	79.12	Pennsylvania.Coordinated Ransomware
Phishing B	100	80	524.3	98.9	79.12		80	79.12	Pennsylvania.Phishing Breach
Remote Ac	100	80	524.3	98.9	79.12		80	79.12	Pennsylvania.Remote Access Exploit
SCADA Ma	100	80	524.3	98.9	79.12		80	79.12	Root Node.Pennsylvania.SCADA Malware
Coordinat	100	80	524.3	98.9	79.12		80	79.12	Root Node.Pennsylvania.Coordinated Ransomware
Phishing B	100	80	524.3	98.9	79.12		80	79.12	Root Node.Pennsylvania.Phishing Breach
Remote Ac	100	80	524.3	98.9	79.12		80	79.12	Root Node.Pennsylvania.Remote Access Exploit
SCADA Ma	100	80	524.3	98.9	79.12		80	79.12	Root Node.Pennsylvania.SCADA Malware
Coordinat	100	80	524.3	98.9	79.12		80	79.12	Root Node.Pennsylvania.Coordinated Ransomware
Phishing B	100	80	524.3	98.9	79.12		80	79.12	Root Node.Pennsylvania.Phishing Breach
Remote Ac	100	80	524.3	98.9	79.12		80	79.12	Root Node.Pennsylvania.Remote Access Exploit
SCADA Ma	100	80	556.3	105	84		80	84	Illinois.SCADA Malware
SCADA Ma	100	15	83.4	15.7	2.35	83.4	5	0.79	Root Node.Illinois.SCADA Malware
SCADA Ma	100	80	524.3	98.9	79.12		80	79.12	SCADA Malware
Coordinat	100	80	524.3	98.9	79.12		80	79.12	Coordinated Ransomware
Phishing B	100	80	524.3	98.9	79.12		80	79.12	Phishing Breach
Remote Ac	100	80	524.3	98.9	79.12		80	79.12	Remote Access Exploit
SCADA Ma	100	15	78.6	14.8	2.22	78.6	5	0.74	Root Node.Pennsylvania.SCADA Malware
Ransomw	100	12	62.9	11.9	1.43	62.9	5	0.6	Root Node.Pennsylvania.Ransomware
Phishing B	100	7	36.7	6.9	0.48	36.7	5	0.35	Root Node.Pennsylvania.Phishing
Remote Ac	100	6	31.5	5.9	0.35	31.3	5.01	0.3	Root Node.Pennsylvania.Remote Access
Ransomw	100	12	66.8	12.6	1.51	66.8	5	0.63	Root Node.Illinois.Ransomware
Phishing B	100	7	38.9	7.3	0.51	38.9	5	0.36	Root Node.Illinois.Phishing
Remote Ac	100	6	33.4	6.3	0.38	33.4	5	0.32	Root Node.Illinois.Remote Access

9. Flow Fault Tree Analysis and Table



Name	Threat	Vulnerabi	Eliminatio	Conseque	Risk Initia	Allocation	Vulnerabi	Risk Redu	Path
Main Sabo	100	12	62.9	11.9	1.43		12	1.43	Pennsylvania.Main Sabotage
Plant Cont	100	11	57.7	10.9	1.2		11	1.2	Pennsylvania.Plant Contamination
Regional f	100	9	47.2	8.9	0.8		9	0.8	Pennsylvania.Regional Flooding
Grid Outa	100	8	41.9	7.9	0.63		8	0.63	Pennsylvania.Grid Outage
Main Sabo	100	12	66.8	12.6	1.51		12	1.51	New Jersey.Main Sabotage
Plant Cont	100	11	61.2	11.5	1.26		11	1.26	New Jersey.Plant Contamination
Regional f	100	9	50.1	9.4	0.85		9	0.85	New Jersey.Regional Flooding
Grid Outa	100	8	44.5	8.4	0.67		8	0.67	New Jersey.Grid Outage
Main Sabo	100	12	66.8	12.6	1.51	66.8	5	0.63	Root Node.New Jersey.Main Sabotage
Plant Cont	100	11	61.2	11.5	1.26	61.2	5	0.58	Root Node.New Jersey.Plant Contamination
Regional f	100	9	50.1	9.4	0.85	50.1	5	0.47	Root Node.New Jersey.Regional Flooding
Grid Outa	100	8	44.5	8.4	0.67	44.39	5.01	0.42	Root Node.New Jersey.Grid Outage
Main Sabo	100	12	62.9	11.9	1.43		12	1.43	Main Sabotage
Plant Cont	100	11	57.7	10.9	1.2		11	1.2	Plant Contamination
Regional f	100	9	47.2	8.9	0.8		9	0.8	Regional Flooding
Grid Outa	100	8	41.9	7.9	0.63		8	0.63	Grid Outage
Main Sabo	100	12	62.9	11.9	1.43	62.9	5	0.6	Root Node.Pennsylvania.Main Sabotage
Plant Cont	100	11	57.7	10.9	1.2	57.7	5	0.55	Root Node.Pennsylvania.Plant Contamination
Regional f	100	9	47.2	8.9	0.8	47.2	5	0.45	Root Node.Pennsylvania.Regional Flooding
Grid Outa	100	8	41.9	7.9	0.63	41.71	5.01	0.4	Root Node.Pennsylvania.Grid Outage